

FAISAL AHMED YOUSIF AL-DROU

Junior AI & ML Engineer

Phone Number: +962780543997 | Email: faisalaldr12327@gmail.com | Address: Amman, Jordan
LinkedIn: linkedin.com/in/faisalaldr12327 | Github: github.com/Faisal-Aldrou

PROFESSIONAL SUMMARY

Artificial Intelligence and Robotics Engineer with practical experience in developing machine learning models, data processing pipelines, and deep learning solutions using Python and modern AI frameworks. Experienced in applied AI projects including forecasting, data analysis, and image classification, with additional experience in computer science and robotics training. Ranked 2nd academically, with the ability to deliver data-driven AI solutions that enhance operational efficiency and support smarter decision-making for the organization.

TECHNICAL SKILLS

Artificial Intelligence & Generative AI:	Machine Learning, Deep Learning, Neural Networks, LangChain Expression Language (LCEL), Multi-Provider Model Routing, Prompt Engineering, Predictive Modeling, RAG & Agentic AI (In Progress)
AI Frameworks & Libraries:	LangChain, RAG Pydantic v2, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna
Data Science & Output Parsing:	Pandas, NumPy, JSON Output Parsing, Feature Engineering, Data Preprocessing, Data Transformation, Data Analysis, Apache Parquet
Web & Middleware Architecture:	Python (Flask), RESTful API Design, Middleware Architecture, Error Handling & Fallback Pipelines, Latency Tracking & Benchmarking
Programming Languages:	Python, SQL, C++, JavaScript (Fetch API / Async), Kotlin, Arduino C
Development Tools & Security:	Git, Linux, Bash, Ubuntu Command Line, Environment Variable Management (python-dotenv), Secret Isolation

PROFESSIONAL EXPERIENCE

Computer Science & Robotics Instructor	Amman, Jordan
Al-Namothaj School for Special Education	
Delivered practical training in computer science and robotics, enabling students to understand programming concepts through hands-on activities.	
- Developed simplified learning materials for complex technical concepts, improving students' understanding and application of programming skills. - Designed interactive technical activities that enhanced students' problem-solving abilities and computational thinking skills.	

TECHNICAL PROJECTS

Autonomous GenAI Sales & RAG Operations Web Application
Technologies: Python (Flask), LangChain (LCEL), ChromaDB, FastEmbed, Pydantic v2, REST API, Groq, Meta Llama, Google Gemini, JavaScript, Gunicorn
Architected an end-to-end Retrieval-Augmented Generation (RAG) assistant capable of multi-turn conversational context, localized dialect processing (Jordanian Arabic), and automated sales pipeline progression.
- Engineered a high-performance semantic search pipeline using ChromaDB and FastEmbed embeddings, utilizing header-aware document chunking to execute sub-second vector similarity retrieval over domain knowledge. - Implemented deterministic lead extraction and classification using Pydantic v2 schemas and LangChain's JsonOutputParser to autonomously detect purchase intent, capture client metadata (name, phone, subscription tier), and persist qualified leads. - Enforced strict closed-domain guardrails within LangChain LCEL pipelines to eliminate hallucinations, prevent unauthorized external URLs, and gracefully handle out-of-scope inquiries. - Built defensive fallback architectures across multi-provider LLMs (Groq/Llama-3.1, GPT-OSS, Google Gemini) to eliminate 429 rate-limit degradation and maintain sub-500ms response latency. - Deployed the production microservice on cloud infrastructure (Render/Gunicorn) with dynamic environment

configuration, health-check monitoring, and continuous 24/7 endpoint availability.

• **ClimaPredict AI – Sub-Seasonal Temperature Forecasting System**

Technologies: Python, LightGBM, XGBoost, CatBoost, Scikit-Learn, Streamlit, Plotly, Optuna, Pandas, NumPy, Joblib

• **New York City Collision Data ETL Pipeline**

Technologies: Python, Pandas, NumPy, JSON, Apache Parquet

Built a data processing pipeline to clean, transform, and organize large-scale collision datasets, improving data quality and preparing it for analysis and insight extraction.

• **Binary Image Classification Using Convolutional Neural Networks (CNN)**

Technologies: Python, TensorFlow, Keras, NumPy, Matplotlib

Developed a deep learning model for automated image classification, enabling pattern recognition through convolutional neural network techniques.

EDUCATION

Artificial Intelligence and Robotics Engineering

Graduation Date: 2024

GPA: 83.3% | Academic Rank: Ranked 2nd in class

TECHNICAL TRAINING & CERTIFICATIONS

- **IBM RAG and Agentic AI Professional Certificate (In Progress)** – Coursera / IBM
 - Completed: "Develop Generative AI Applications" (LangChain, LCEL, Pydantic, Flask)
 - Completed: "Build RAG Applications" (LlamaIndex, RAG, Embeddings, Gradio, Document Management)
- **Data Science and Artificial Intelligence Training** – SHAI for AI
- **Practical Training Program** – Infotel
- **Artificial Intelligence and Machine Learning in Medical Applications** – Course The Hope International
- **Python for Machine Learning Certificate** – Pro Vision

LANGUAGES

Arabic: Native

English: Advanced

Japanese: Intermediate